**27/03/2023 — 2 hours** — Optimization and Operational Reasearch

- Calculator are allowed
- Provide all the calculation details

# PART 1 - Convexity

Study the convexity of the following functions

**Question 1—**

$$f(x,y) = -\log(\frac{1}{4x+y}), \quad f(x,y) = \exp(x-2y) - x \text{ and } \quad f(x,y,z) = 2x^2 + 2xy + 2y^2 + z^2$$

**Question 2—**A function is convex if its restriction to every line is convex. For a function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $f$ is convex if $g(t) = f(z + tv)$ is convex with $t \in \mathbb{R}$ for all $z, v \in \mathbb{R}^2$. Use this property to study the convexity of

$$f(x,y) = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

# PART 2 - Unconstrained Optimization

$$\arg\min_x f(x) = ax_1^2 + 2x_1x_2 + x_2^2$$

**Question 3—**Provide the gradient of $f$ in the form (i.e. explicit the value of $B$) :

$$\nabla_x f(x) = Bx$$

**Question 4—**For which value of $a$ is the function $f$ convex ?

**Question 5—**Apply the gradient descent algorithm with optimal step for a = 2 to minimize $f$ starting at (2,1), with

# PART 3 - Constrained Optimization

$$g_\eta(x) = \eta x_1^3 + \frac{1}{2}(2x_1^2 + x_2^2 + x_1x_2) - 6x_1 + 2x_2$$

**Question 6—**Study the convexity of $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$, which is parameterized by a value $\eta \in \mathbb{R}$.

**Question 7—**give the solution of Euler's equations for all value of $\eta \notin \{-2\sqrt{2}, 2\sqrt{2}\}$. What happens when $\eta = -2\sqrt{2}$ or $\eta = 2\sqrt{2}$

**Question 8—**Give the nature of the previous extrema (i.e. say if it is local/global minimum/maximum of the function). The nature of the extremum depends on $\eta$

# PART 4 - Constrained Optimization

$$\arg\min_x f(x) = 2x^2 - 4x - 2$$
$$\text{s.t.} \quad -4x \leq -2$$

**Question 9—**Write the Lagrangian of this problem.

**Question 10—**What is the dual function ? Deduce the lower bound of the optimum.

**Question 11—**Does strong duality holds ? Prove it if needed.

**Question 12—**Find the optimal values with respect to KKT conditions